

Evaluating Generative Models for Cloud Removal in Satellite Imagery

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1 Problem Definition & Use Case

1.1 Problem Definition: Cloud Removal Problem

Optical remote sensing in southern China is severely affected by persistent cloud cover, with annual cloud frequency exceeding 80% in some areas. This creates "data islands," making critical time-series monitoring—such as crop growth cycles, illegal logging, or disaster response—impossible. This monitoring is necessary because these cloud-covered regions are also China's agricultural areas and lower-income regions. If these areas lose natural disaster monitoring, it often leads to more severe economic impacts—because farmers in these regions depend on agriculture for income growth.

In recent years, generative AI models have brought revolutionary breakthroughs to remote sensing image reconstruction. These models can learn the spatial texture features of optical imagery, thereby "synthesizing" missing surface information in cloud-obscured areas.

When developing generative models for cloud occlusion restoration, this study focuses on exploring their performance boundaries: quantifying the decay relationship between cloud coverage ratio and reconstruction data reliability, to determine at what occlusion threshold the model will produce physical distortions.

1.2 Application Scenario: Landmark Monitoring in Cloudy and Rainy Regions

This image completion capability enables many urban planning problems to be solved more clearly. For example, when conducting land use change monitoring, urban expansion assessment, or green space coverage statistics in cloudy southern regions, data gaps caused by cloud cover in the past often prevented the formation of complete spatiotemporal analysis chains.

The findings and outputs of this project will inform critical financial and policy decisions:

1. Subsidy Distribution: Allowing agricultural agencies to verify the spatial extent of planted crops in persistently cloudy regions to ensure accurate and equitable distribution of farming subsidies.
2. Economic & Yield Assessment: Enabling stakeholders to estimate early-season crop vitality and potential harvest yields, informing poverty alleviation strategies and market interventions.

2 Technical Justification

2.1 Method Selection and Rationale: Why Generative Models

Since completing satellite imagery is a problem of filling in missing information, the model first needs to learn the conditional probability distribution of optical imagery for a given region. Under cloud cover conditions, observational data is incomplete, and the model needs to infer the occluded areas based on information from visible regions.

In this process, the model primarily learns two types of key features: first, the spatial texture structure (spatial patterns) of the land surface, such as the regular arrangement of farmland, the grid patterns of cities, or the canopy distribution of forests; second, spatial contextual dependencies (contextual reasoning), which means using information from visible areas around the clouds to infer the content of occluded regions.

Because cloud removal is an image inpainting problem, compared to classification or segmentation methods, generative models can recover continuous spatial information in areas where observations are completely missing, rather than only outputting discrete labels. For example, classification tasks can only identify land cover types and cannot provide spatial details; therefore, generative approaches are a necessary pathway to solve this problem.

At the same time, this study’s baseline selects the traditional multi-temporal fusion method—that is, without using deep learning methods, completing missing images by utilizing multi-temporal, multi-frame imagery.

This study employs diffusion-based frameworks and GANs to complete image inpainting under cloud cover conditions, and attempts to compare and explain: Does the generative model have advantages compared to traditional multi-temporal fusion methods? What are the advantages? And at what boundaries does each method fail?

2.2 Common Failure Modes of Generative Models

Hallucination under high cloud coverage

When the cloud coverage proportion is high, the generative model will may present outcomes that are visually plausible but inconsistent with the actual land surface.

Inability to predict satellite optical imagery after sudden events

Generative models are essentially based on learning the statistical distribution and structural patterns of the land surface from historical data, so their predictions depend on existing patterns and features. When facing sudden events (such as floods), the model struggles to generate imagery reflecting the current true state, and is more inclined to output a "historical state."

3 Methodological Precedent

3.1 Precedent Methods

Traditional Remote Sensing Cloud Removal

Before the emergence of generative models, cloud removal in the remote sensing field primarily relied on multi-temporal compositing methods.

Multi-temporal methods utilize multiple images of the same area acquired at different times, reconstructing complete images by filtering cloud-free pixels. For example, Landsat-based compositing methods (Roy et al. 2010 Landsat compositing) generate cloud-free images through observations within a temporal window.

The advantage of this method is its ability to perform reconstruction based on real observational data. However, its limitations are also very obvious: if a region is under persistent cloud cover, there will be a lack of cloud-free observations for all pixels in that area over a period of time.

We will borrow this method to build our non-deep-learning baseline.

GAN-based Cloud Removal Methods (Enomoto et al., 2017; Isola et al., 2017)

Generative Adversarial Networks (GANs), first proposed by Goodfellow et al. (2014), were an earlier method used to solve image generation problems. In the remote sensing and geographic information research field, early studies of cloud removal adopted GAN models to generate cloud-free images from cloudy images. Such methods typically construct paired data (cloudy / cloud-free pairs) based on Landsat or Sentinel-2 satellite imagery.

GAN models learn a conditional mapping function from cloudy to cloud-free images, where the generator is responsible for generating de-clouded images, while the discriminator distinguishes between generated images and real cloud-free images, forming an adversarial optimization process. The model attempts to capture spatial structural features and texture patterns, such as the geometric arrangement

of farmland, the grid structure of cities, and the continuity of vegetation coverage, thereby generating content consistent with the surrounding context in cloud-occluded regions.

Application of Diffusion Models in Image Inpainting (Diffusion-based Inpainting)

Diffusion Models, proposed in recent years, provide a more stable and expressive modeling framework for image generation and inpainting tasks. Typical work such as the Denoising Diffusion Probabilistic Model (DDPM) proposed by Ho et al. (2020) achieves data distribution modeling by progressively adding noise to data and learning the reverse process. In image inpainting tasks, the RePaint method proposed by Lugmayr et al. (2022) further introduces mask-guided sampling, enabling the model to perform high-quality reconstruction of missing regions conditioned on known areas.

In remote sensing applications, this method can naturally extend to cloud mask-based conditional generation problems: treating cloud-covered regions as missing areas, and gradually generating complete images through the diffusion process while preserving observational information in cloud-free areas. This enables the model to fully utilize spatial contextual information, alleviating to some extent the information loss problem of traditional methods under high cloud cover conditions.

3.2 Research Gap

Existing research mainly focuses on "whether complete images can be generated," and less systematically explores whether generated results are reliable and under what boundary conditions generated results are reliable. Especially under varying cloud coverage ratios, how the performance of different methods decays and at what threshold significant distortion begins to occur still lacks clear analysis.

Therefore, this paper does not simply propose a new generation method, but rather places the above three types of methods within a unified framework for comparison, focusing on their performance boundaries: analyzing the relationship between cloud coverage ratio and reconstruction reliability, thereby identifying under what conditions different methods can be trusted, and under what conditions they may produce misleading results.

4 Data Plan: Synthetic Cloud Benchmark for Performance Boundary Analysis

Since the core objective of this study is to quantify the relationship between cloud coverage ratio and reconstruction reliability, the experimental data needs to simultaneously satisfy two conditions: first, having real cloud-free imagery (ground truth); second, being able to precisely control cloud coverage ratio. However, in high-cloud-frequency regions such as southern China, real cloud-free imagery is difficult to obtain, so this study adopts a strategy of "low-cloud regions + synthetic cloud generation" to construct the dataset.

Therefore, this study selects low-cloud-coverage regions in northern China (such as Beijing and surrounding areas) as the data source to ensure stable acquisition of cloud-free imagery as ground truth.

Dataset: Sentinel-2 10m resolution optical remote sensing imagery (Beijing and surrounding areas)

Total data volume: 2000 cloud-free images as ground truth data (randomly split into 80% training set, 20% test set)

Cloud coverage ratios: 5%, 10%, 30%, 50%, 70%, to simulate different scenarios from low occlusion to high occlusion; (generated by randomly placing cloud patches on ground truth cloud-free images)

5 Modeling Approach

Baseline model: multi-temporal compositing (reconstruction by selecting cloud-free pixels based on multi-temporal imagery)

Model A: GAN-based model (learning the mapping from cloudy to cloud-free imagery)

Model B: Diffusion-based inpainting model (progressive generation and completion under cloud mask constraints)

Preprocessing: Standardize Sentinel-2 multispectral imagery and generate corresponding cloud masks as input conditions

Feature engineering: Use RGB and NIR bands as input feature values, with RGB images as output results

6 Evaluation Strategy

Qualitative Analysis

Conduct visual comparisons of reconstruction results in typical regions, focusing on structural recovery under high cloud coverage conditions. Pay special attention to whether there are cases of "visually plausible but inconsistent with actual land surface" (hallucination), and performance differences of models in complex land surfaces (such as farmland boundaries and urban structures).

Quantitative Evaluation

At the pixel level, use PSNR and SSIM to measure the difference between generated imagery and ground truth, to evaluate reconstruction accuracy and structural consistency. Additionally, introduce NDVI error as a remote sensing-related metric to assess the reliability of model reconstruction in vegetation areas.¹

In the experiments, cloud coverage ratio is used as the core variable, with each metric calculated separately at different occlusion levels (such as 5%, 10%, 30%, 50%, 70%), and cloud coverage ratio-error curves plotted to analyze the performance degradation trends of different methods and identify their performance boundaries.

Cited Work

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¹*PSNR (Peak Signal-to-Noise Ratio) is used to measure pixel-level reconstruction error between generated and real imagery. SSIM (Structural Similarity Index) is used to evaluate perceptual similarity of images in terms of brightness, contrast, and spatial structure. NDVI (Normalized Difference Vegetation Index) is calculated based on red and near-infrared (NIR) bands to reflect vegetation information, and is used in this study to assess the accuracy of land surface recovery in vegetation areas of reconstruction results.